

STATEMENT OF WORK

NATURE OF PROJECT:	ML Challenge
PROJECT REFERENCE NAME:	<i>Thesis challenge</i>

Description of Services to be completed by:

1. Project: ***Thesis challenge***

An approach to a possible solution to the thesis topic

Human resource needs

- Python developer
- Data Scientist
- MLOps

The project should be completed in several stages:

- **Stage 1** Download data set
 - **Data Set Information**
- **Stage 2** Will create a models
 - **Preprocessing**

This endpoint will receive as input:

- Attribute
- Domain

This endpoint will receive as input data set result in a data frame

- Data info
- Data describe
- Data values and counts
- Nulls and others characters

- Data normalization

- **Models**

This endpoint will receive as input parameters

- Construction models
- End Date (date time)
- Taring model
- Cross validation
- Test model

- **Metrics**

- Precision
- Recall
- F1-Scores
- Confusion Matrix
- ROC

- **Stage 3 Create a pipeline to MLOps**

- The MLOps part will be done with mlflow performing the following tasks
 - Log metrics
 - Model signatures
 - Save the plot and log it as an artifact
 - Tracking url (localhost)
 - Run MLOps

Section 2.

Other Notes and Considerations:

- Use a personal account per team to connect to git hub
- Best Practices matters
- Code should be continuously uploaded to a list of users will be provided to be added as collaborators to the repo.
- Validate input parameters of all requests
- Exception handling is excepted

Section 3.

Documentation

Expected documentation should be in English and will include:

- Web Scraping
- Model Construction
- MLOps
- Instructions on how to pull the code from github, compile, run the solution and post a request. These instructions are targeted to technical.

Extra documentation (nice to have):

- Architecture Diagrams